Global Fundamental-Parameter Mapping of Lunar Elemental Abundances from Chandrayaan-2 CLASS XRF Data with Cross-Modal Super-Resolution

Smarak Kanjilal, Subarno Maji, *and* Paidi Geetesh Chandra

***Abstract*—Mapping the elemental composition of the Moon at high spatial resolution is critical for understanding its geological evolution and resource potential. We present a complete pipeline for lunar elemental abundance mapping from Chandrayaan-2 Large Area Soft X-ray Spectrometer (CLASS) X-ray fluorescence (XRF) data. Raw Level-1 spectra are preprocessed with empirical nightside-background subtraction and a stationary-wavelet-transform plus Savitzky–Golay denoiser, then fitted with a physics-based fundamental-parameter (FP) forward model – including primary and secondary fluorescence – using whole-spectrum Trust-Region-Reflective (TRF) optimisation, rather than region-of-interest line-ratio sums. Per-footprint element/Si ratios are projected onto a global $\sim 0.33^\circ$ grid by centre-point binning with Gaussian smoothing and gap infill. Applied to 1,056,619 dayside spectra from 24 monthly archives (December 2020–May 2026), the pipeline attains 99.0% surface coverage at 1° resolution. We further introduce a cross-modal, optical-guided super-resolution stage that couples the XRF maps to a co-registered optical mosaic through ridge-regularised regression and frequency-domain fusion. On the reliable equatorial-to-mid-latitude band the two clearest tracers behave as expected geologically: Al/Si is the most robust positive correlate of surface brightness ($\rho = +0.43$, bright feldspathic highlands) and Mg/Si the most robust negative correlate ($\rho = -0.39$, dark mafic maria) – moderate correlations, but the most consistently signed among the six mapped elements. Validation in element/Si ratio space against nine Apollo and Luna landing sites shows the fitted Al/Si ratio tracks ground truth at $r = +0.74$, indicating the pipeline recovers a real compositional signal. Fitted concentrations are reported as relative scale factors pending absolute calibration.**

***Index Terms*—Lunar remote sensing, X-ray fluorescence spectroscopy, elemental abundance mapping, Chandrayaan-2, fundamental-parameter modeling, cross-modal data fusion, super-resolution.**

I. Introduction

A. Background

THE composition of the Moon's surface is central to understanding its geological and evolutionary history. Dark basaltic plains mark past volcanism and record the Moon's thermal and magmatic evolution; the surface itself is a mixture of crustal materials, impact-generated deposits, and space-weathering products from solar-wind interaction. Remote sensing has long been used to map global elemental abundances, whose spatial distribution remains central to understanding lunar formation and evolution.

All authors are Independent Researchers (e-mail: smarakkanjilal@gmail.com; subarnomaji@kgpian.iitkgp.ac.in; paidigeeteshchandra@gmail.com). ORCID: 0009-0005-4361-3119 (S. Kanjilal); 0009-0006-4698-6882 (S. Maji); 0009-0008-1153-3079 (P. G. Chandra).

Smarak Kanjilal, Subarno Maji, and Paidi Geetesh Chandra contributed equally to this work. (Corresponding author: Paidi Geetesh Chandra.)

B. Motivation and Objectives

This study generates high-resolution lunar maps from X-ray fluorescence (XRF) line intensity ratios. Ratioing minimises the influence of external factors such as solar flux and viewing geometry, yielding an independent geochemical view of the surface. The resulting maps characterise compositional heterogeneity relevant to both scientific investigation and in situ resource utilisation, and are intended to contribute toward the next generation of lunar mapping missions.

C. XRF Spectroscopy Overview

X-ray fluorescence (XRF) spectroscopy is a well-established technique for characterising the Moon's surface composition [1], [2]. The Chandrayaan-1 X-ray Spectrometer (C1XS) provided early geochemical insight into regions such as the southern highlands, validated against laboratory experiments and Monte Carlo simulations. The Chandrayaan-2 Large Area Soft X-ray Spectrometer (CLASS) then improved spatial resolution and sensitivity [3], with demonstrated spatial resolutions of 150 to 15 km. Challenges remain from factors affecting XRF line intensities – incident solar spectrum, matrix effects, particle size, and geometry (angle of incidence, phase angle) – particularly in nadir-viewing configurations like CLASS.

D. Previous Work

Several lunar missions have advanced understanding of surface composition through direct sampling and remote sensing. The Lunar Prospector gamma-ray spectrometer (GRS) produced the first global elemental maps [4] at coarse (tens to hundreds of km) resolution, mapping thorium and potassium well but struggling with Mg, Al, and Si owing to moderate spectral resolution and detector background. Chandrayaan-1 and -2 (C1XS, CLASS) subsequently improved spatial resolution and sensitivity for XRF line ratios such as Mg/Si and Al/Si. Lunar soil samples from Apollo, Luna, and Chang'E-5, together with rover-based in situ measurements, provide complementary ground-truth references.

Machine learning has further advanced lunar composition analysis. LSCC reflectance spectra have been classified with Support Vector Machines and penalised logistic regression to determine particle size and maturity [5]. Particle Swarm Optimisation coupled with XGBoost (PSO-XGBoost) has modelled oxide abundances against geochemical indicators to yield high-resolution global oxide maps [6], and Bayesian-optimised XGBoost has been applied to landing-site safety assessment, outperforming CNNs and Random Forests on feature importance [7]. Chang'E-5 data analysed with 1d-CNN models have produced detailed oxide distribution maps [8], underscoring the growing role of deep learning in lunar surface analysis; unlike these learned methods, the fundamental-parameter approach adopted here requires no training data, at the cost of the simplifications noted in Sections III-C and VIII.

II. Spectral Preprocessing

A. Data acquisition

The X-ray spectral data used in this study were acquired from the Chandrayaan-2 Large Area Soft X-ray Spectrometer (CLASS). The dataset was obtained from the PRADAN (Policy-based Data Retrieval, Analytics, Dissemination, and Notification system) portal hosted by the Indian Space Science Data Centre (ISSDC), Indian Space Research Organisation (ISRO). The data comprise 24 non-contiguous monthly archives spanning December 2020 to May 2026, totalling 1,515,239 raw Level-1 exposures.

CLASS performs X-ray fluorescence (XRF) spectroscopy using Swept Charge Devices (SCDs) to detect characteristic X-rays from the lunar surface, operating in the soft X-ray band (0.8–15 keV) to map major rock-forming elements including Mg, Al, Si, Ca, Ti, and Fe. Raw files were accessed in FITS format, containing photon counts, exposure times, and detector gain information.

B. Data Preprocessing

A custom pipeline processes the raw spectra and extracts elemental fluxes in four stages: reference calibration, background subtraction, signal denoising, and flux integration.

1. Reference Energy Calibration: Accurate reference energies for the target elements (O, Mg, Al, Si, Ca, Ti, Fe) were established using the xraydb database. The K-shell emission lines were queried for each element. Priority was given to the $K_{\alpha 1}$ line; however, if unavailable, the generic K_{α} line was selected. These reference energies (in keV) served as the central centroids for the Region of Interest (ROI) calculations.

2. Background Subtraction: As no dedicated background file ships with this CLASS data release, a reference background spectrum was constructed empirically by averaging the night-side exposures (SOLARANG $\geq 90°$) within each monthly batch, which receive negligible solar-driven fluorescence and so capture the instrumental and cosmic background. Both spectra were normalised by exposure time to count rates (counts/s), the background rate was subtracted, and a non-negativity constraint was applied to the result.

3. Signal Denoising: To improve the signal-to-noise ratio, a multi-stage denoising approach was utilized. First, a Stationary Wavelet Transform (SWT) was applied to the background-subtracted signal using the Symlet-8 (sym8) wavelet at decomposition level 2 [9]. Soft thresholding was applied to the detail coefficients, with the threshold dynamically determined from the signal's noise variance (σ). Following wavelet reconstruction, the spectrum was smoothed using a Savitzky–Golay filter (window length 7, polynomial order 3) [10] to preserve spectral line shapes while reducing high-frequency noise.

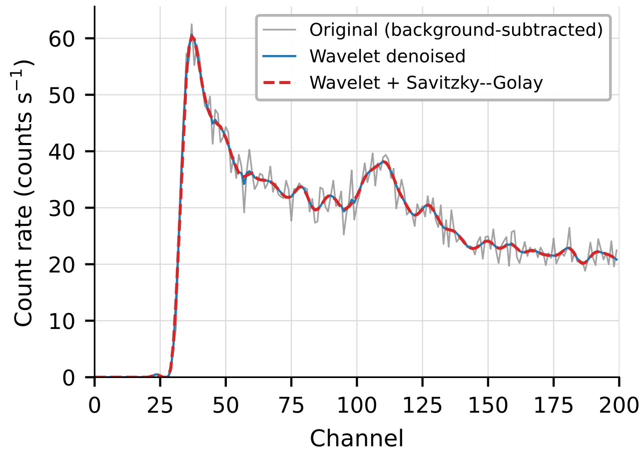

Fig. 1. Spectral preprocessing pipeline (Section II): raw count-rate spectrum, background-subtracted spectrum, and the final wavelet+Savitzky–Golay denoised spectrum, overlaid for one representative dayside exposure. Dashed vertical lines mark the Kα line energies of Mg, Al, Si, Ca, Ti, and Fe.

4. Flux Integration: Unlike traditional ROI-sum XRF analysis, the denoised spectrum is not collapsed into a single integrated count per element here. Instead, the full background-subtracted, denoised, non-negativity-constrained count-rate array is retained across all N_{ch} channels and passed directly as the measured spectrum $\mathbf{S}_{\text{meas}}$ to the whole-spectrum fundamental-parameter fit (Section IV). This is necessary because the Kα/Kβ lines of adjacent light elements (e.g. Mg, Al, Si) partially overlap once convolved with the detector response (Section III-E); a discrete ROI sum would conflate these, whereas the whole-spectrum TRF fit can disentangle them using the full model shape, including secondary-fluorescence cross-terms (Section III-C).

Figure 1 illustrates the three-stage preprocessing pipeline on one representative dayside exposure.

III. Fundamental-Parameter Forward Model

Quantitative analysis of XRF spectra can follow either an *empirical* calibration approach or a *fundamental-parameter* (FP) approach. The FP method derives fluorescence intensities directly from the physics of X-ray interaction with matter, avoiding the need for matrix-matched calibration standards [11]–[13]. This section describes the FP forward model used in the pipeline.

A. Atomic Data and Line Setup

1) Line Energies and Radiative Rates: For each element (identified by atomic number Z), the model retrieves K-series emission line energies and radiative rates from the `xraylib` database [14]:

$$E_\ell : \text{line energy for transition } \ell, \tag{1}$$

$$p_\ell : \text{radiative rate for transition } \ell. \tag{2}$$

The lines included are Kα_1 and Kβ_1. L-series lines are supported in the framework but are not employed for the light-to-medium-Z elements considered here.

Lines are grouped by *type* (Kα, Kβ, Lα, Lβ), and for each type a *mean energy* is computed:

$$\bar{E}_t = \frac{1}{|\mathcal{L}_t|} \sum_{\ell \in \mathcal{L}_t} E_\ell, \tag{3}$$

where $\mathcal{L}_t$ is the set of lines belonging to type t.

2) Mass Attenuation Coefficient: The total mass attenuation coefficient $\mu(E)$ [cm^2/g] at a given energy E is obtained from tabulated cross-sections:

$$\mu_i(E) = \sigma_{\text{total}}(Z_i, E), \tag{4}$$

which includes photoelectric, coherent (Rayleigh), and incoherent (Compton) contributions. This quantity is used both for primary intensity calculations and for secondary fluorescence cross-terms.

3) Jump Factor: At an absorption edge of shell s, the mass attenuation coefficient exhibits a discontinuity characterised by the *jump ratio* r_s. The *jump factor* is defined as:

$$\boxed{J_s = 1 - \frac{1}{r_s}} \tag{5}$$

The jump factor represents the fraction of total photoelectric absorption attributable to shell s ionisation. Just below the K-edge, only L, M, $\ldots$ shells contribute; at the edge, K-shell ionisation becomes energetically accessible, causing a discontinuous increase in the cross-section.

4) Fluorescence Yield: The fluorescence yield ω_s is the probability that a vacancy in shell s is filled radiatively (X-ray emission) rather than by Auger emission. For the elements in the model, ω increases with atomic number (e.g. $\omega_{\text{K}}(\text{Mg}) \approx 0.02$, $\omega_{\text{K}}(\text{Fe}) \approx 0.33$).

5) Elemental Constant: Combining the three fundamental quantities above, the *elemental constant* for element i at emission line ℓ (belonging to shell s) is:

$$\boxed{K_{i,\ell} = J_s^{(i)} \cdot \omega_s^{(i)} \cdot p_\ell^{(i)}} \tag{6}$$

The shell is determined from the line type: Kα or Kβ lines correspond to the K shell ($s = 0$), while Lα or Lβ lines correspond to the L shell ($s = 1$).

B. Primary Fluorescence Intensity

The primary fluorescence intensity for element i at emission line ℓ is proportional to its mass attenuation coefficient evaluated at the emission line energy:

$$\boxed{I_{i,\ell}^{(\text{P})} = \mu_i(E_\ell)} \tag{7}$$

C. Secondary Fluorescence Intensity

Secondary fluorescence occurs when a characteristic photon emitted by element j is absorbed by element i, causing additional fluorescence of element i. This inter-element enhancement effect is particularly important in multi-element lunar samples where the mutual excitation between neighbouring elements can significantly alter the observed intensities.

The secondary intensity contribution to line ℓ of element i (the analyte) is:

$$\boxed{I_{i,\ell}^{(\text{S})} = \sum_{\substack{j=1 \\ j \neq i}}^{N} \sum_{m \in \mathcal{L}_j} \mu_j(E_m) \cdot \mu_i(E_m) \cdot K_{i,\ell} \cdot \beta_{j,m}} \tag{8}$$

where:

- $\mu_j(E_m)$: mass attenuation coefficient of element j at the energy of its own emission line m.
- $\mu_i(E_m)$: mass attenuation coefficient of element i (the analyte) at the energy of element j's emission line m, quantifying how strongly i absorbs j's fluorescence.
- $K_{i,\ell}$: the elemental constant for the analyte line (Eq. (6)).
- $\beta_{j,m} \in [0, \infty)$: an adjustable enhancement factor for element j, line m (default value 1.0), accounting for geometry-dependent effects not captured by the simplified formula.

The outer sum runs over all elements $j \neq i$; the inner sum runs over all available emission lines m of element j. For simplicity, the geometric terms (ψ, ϕ) of the full Sherman (1955) secondary- fluorescence formulation are absorbed into the scale factor s (Eq. (17)), which is fitted per spectrum.

D. Total Fluorescence Intensity

The total fluorescence intensity for element i at line ℓ, combining primary and secondary excitation, is:

$$\boxed{I_{i,\ell}^{(\text{tot})} = C_i \cdot \left[I_{i,\ell}^{(\text{P})} + I_{i,\ell}^{(\text{S})}\right]} \tag{9}$$

where C_i is the concentration (weight fraction) of element i. The emitted intensity is thus proportional to both the analyte abundance and the strength of the excitation channels.

E. Gaussian Detector Response

An energy-dispersive detector has finite energy resolution that broadens each monochromatic emission line into a peak with an approximately Gaussian profile. The detector-response function for a line at energy E_ℓ is:

$$\boxed{G(E;\, E_\ell,\, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(E - E_\ell)^2}{2\sigma^2}\right]} \tag{10}$$

where σ is the standard deviation, related to the full width at half maximum (FWHM) by:

$$\text{FWHM} = 2\sqrt{2\ln 2}\,\sigma \approx 2.355\,\sigma. \tag{11}$$

F. Synthetic Spectrum Assembly

1) Channel Discretisation: The energy axis is discretised into detector channels:

$$E_k = k \cdot \Delta E, \qquad k = 0, 1, \ldots, N_{\text{ch}} - 1, \tag{12}$$

where N_{ch} is the number of channels and ΔE is the energy per channel:

$$\Delta E = \begin{cases} 0.0135 \text{ keV/channel} & \text{for } N_{\text{ch}} = 2048, \\ 0.0277 \text{ keV/channel} & \text{for } N_{\text{ch}} = 1024. \end{cases} \tag{13}$$

TABLE I
BOUND CONSTRAINTS USED IN THE TRF FIT FOR EACH PARAMETER IN $\boldsymbol{\theta}$ (EQ. (18)). THE CONCENTRATION BOUND OF 200 IS NOT A WT% LIMIT: C_i ARE UNITLESS RELATIVE SCALE FACTORS, NOT CALIBRATED TO WT% (SECTION IV-A).

Parameter	$\theta_{\min}$	$\theta_{\max}$
C_{Fe}, C_{Al}, C_{Mg}, C_{Si}, C_{Ca}, C_{Ti}, C_{O}	0	200
s (scale)	10^{-8}	1
σ (keV)	0.01	0.5

2) Windowed Gaussian Convolution: For each element i and each line ℓ, the Gaussian convolution is evaluated only within a truncation window of $\pm\sigma$ around the line centre $\bar{E}_t$ (the mean energy of the line type). The bin boundaries are:

$$k_{\mathrm{start}} = \max\left(0,\ \left\lfloor \frac{\bar{E}_t - \sigma}{\Delta E} \right\rfloor\right), \tag{14}$$

$$k_{\mathrm{end}} = \min\left(N_{\mathrm{ch}},\ \left\lceil \frac{\bar{E}_t + \sigma}{\Delta E} \right\rceil\right). \tag{15}$$

The number of evaluation points within the window is:

$$N_{\mathrm{bins}} = \left\lfloor \frac{2\sigma}{\Delta E} \right\rfloor + 1. \tag{16}$$

3) Full Spectrum Formula: The total synthetic spectrum is the sum over all elements and all their lines:

$$\boxed{\begin{gathered} S_{\mathrm{model}}(E_k) = s \sum_{i=1}^{N} \sum_{\ell \in \mathcal{L}_i} I_{i,\ell}^{(\mathrm{tot})} \cdot G(E_k;\ \bar{E}_t,\ \sigma) \\ \text{for } k \in [k_{\mathrm{start}},\ k_{\mathrm{end}}) \end{gathered}} \tag{17}$$

where s is a global scale factor and $\bar{E}_t$ is the mean energy of the line type containing ℓ.

IV. SPECTRAL FITTING PROCEDURE

The forward model described in Section III is fitted to each measured spectrum to determine the elemental concentrations and instrument parameters. The nine free parameters to be determined are:

$$\boldsymbol{\theta} = \left(C_{\mathrm{Fe}},\ C_{\mathrm{Al}},\ C_{\mathrm{Mg}},\ C_{\mathrm{Si}},\ C_{\mathrm{Ca}},\ C_{\mathrm{Ti}},\ C_{\mathrm{O}},\ s,\ \sigma\right) \in \mathbb{R}^9, \tag{18}$$

comprising the weight-fraction concentrations of seven elements, a global scale factor s, and the detector resolution parameter σ.

The fit solves the following bound-constrained least-squares problem:

$$\min_{\boldsymbol{\theta}} \left\|\mathbf{S}_{\mathrm{meas}} - \mathbf{S}_{\mathrm{model}}(\boldsymbol{\theta})\right\|_2^2 \quad \text{subject to} \quad \boldsymbol{\theta}_{\min} \le \boldsymbol{\theta} \le \boldsymbol{\theta}_{\max}, \tag{19}$$

where the element-wise bounds $\boldsymbol{\theta}_{\min}$ and $\boldsymbol{\theta}_{\max}$ are given in Table I, and the residual vector is $\mathbf{r}(\boldsymbol{\theta}) = \mathbf{S}_{\mathrm{meas}} - \mathbf{S}_{\mathrm{model}}(\boldsymbol{\theta})$.

At each optimiser iteration, the forward model extracts the current concentrations, scale, and resolution from the parameter vector; computes the primary and secondary fluorescence intensities for every element and emission line; convolves each line contribution with the Gaussian detector-response function within the appropriate windowed channel range; and sums all contributions scaled by s to produce the synthetic spectrum $S_{\mathrm{model}}(\boldsymbol{\theta})$.

The minimisation employs the Trust-Region Reflective (TRF) algorithm [15], which is well-suited for bound-constrained nonlinear least-squares problems. The TRF method iteratively constructs a local quadratic model of the objective function within a trust region, solving a sequence of constrained sub-problems to converge on the optimal parameter vector $\boldsymbol{\theta}^*$. The pipeline is designed for batch processing of large spectral datasets: each spectrum from the catalogue is a 1D array of photon counts with $N_{\mathrm{ch}} = 2048$ channels, and multiple spectra are processed in parallel to handle the large volume of CLASS observations efficiently.

A. Units Caveat

The concentration parameters C_i recovered by the fit (Eq. (18)) are relative scale factors internally consistent with the primary/secondary fluorescence formulation of Section III, not concentrations calibrated to weight percent (wt%): no matrix-matched standards or absolute-intensity calibration were used. C_i values throughout this paper (Tables II to IV and Sections V and VI) should therefore be read as internally consistent relative abundances, informative for spatial pattern and element-to-element ratios but not calibrated wt% figures. A one-point or linear calibration against a reference dataset (e.g. Apollo geochemistry, Section VII-D) is a necessary next step before absolute wt% maps can be produced; Section VII-D validates in ratio space (element/Si) for exactly this reason.

V. SPATIAL ABUNDANCE MAPPING

Once elemental concentrations have been determined for each spectrum, the fitted values are projected onto a geographic grid to produce spatial abundance maps.

A. Ratio Quantity and Outlier Handling

Rather than mapping the raw fitted concentration C_i, we map the *element/silicon ratio* C_i/C_{Si}. Because each fit's overall amplitude is absorbed into the per-spectrum scale factor s (Section IV-A), which varies with solar flux and viewing geometry from exposure to exposure, ratioing to a common denominator (Si, a ubiquitous major element) cancels that nuisance amplitude and yields a quantity that reflects surface composition far more faithfully – the standard practice for lunar XRF line-ratio mapping (e.g. Mg/Si, Al/Si). Before mapping, the per-footprint ratio distribution is cleaned in two steps: a $\log(1+x)$ transform followed by a $1.5\times$ interquartile-range cut removes extreme outliers (a handful of near-zero C_{Si} fits produce non-physical ratios), and a high-percentile cap trims the residual upper tail so the colour scale reflects the bulk of the distribution.

B. Rasterisation and Smoothing

Each observation carries a quadrilateral footprint with vertices $(\mathrm{lat}_k, \mathrm{lon}_k)$, $k = 0, \ldots, 3$. The footprint is reduced to its geometric centre $(\overline{\mathrm{lat}}, \overline{\mathrm{lon}})$ and binned onto an equirectangular

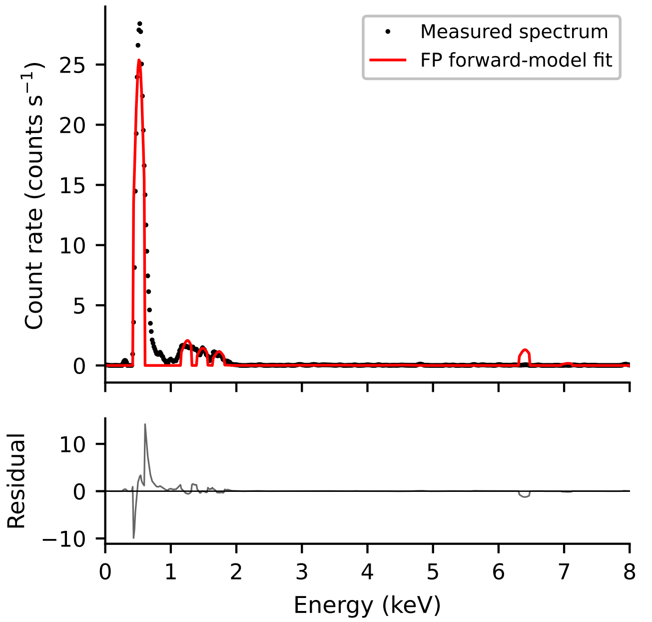

Fig. 2. A representative measured spectrum (black dots) with its best-fit fundamental-parameter model (red line) and fit residuals (bottom panel), zoomed to 0–8 keV where all modeled line energies lie. The spectrum shown was selected as the one closest to the median TRF cost among the top-flux quartile of successfully fitted spectra (i.e. the higher-signal population the maps and super-resolution stage are actually built from, Section V), so the line structure – rather than a near-featureless low-count exposure – is visible. Converged concentrations for this spectrum: Fe 0.00484, Al 0.04714, Mg 0.08601, Si 0.02846, Ca 0.00009, Ti 0.00008, O 13.586 (relative units, Section IV-A); O's value is orders of magnitude larger because its soft, poorly-resolved Kα line (0.52 keV) causes its parameter to absorb low-energy continuum rather than track a resolved line, so it is not used as a quantitative abundance anywhere in this paper. Scale $s = 2.64 \times 10^{-5}$, $\sigma = 0.083$ keV, cost 441.5.

grid of width N and height $N/2$ (default $N = 1080$, i.e. $\sim 0.33°$ per pixel) at indices

$$j = \left\lfloor \frac{\bar{\text{lon}} + 180}{360} N \right\rfloor, \qquad i = \left\lfloor \frac{\bar{\text{lat}} + 90}{180} \frac{N}{2} \right\rfloor. \tag{20}$$

Each cell accumulates the sum of contributing ratio values and a count; their quotient gives the per-cell mean. To suppress per-pixel sampling noise while preserving structure, both the summed-value grid and the occupancy mask are convolved with a Gaussian ($\sigma = 1.1$ px) and the smoothed value is renormalised by the smoothed occupancy:

$$\tilde{\mathbf{A}}(x, y) = \frac{(G_\sigma * \mathbf{V})(x, y)}{(G_\sigma * \mathbf{W})(x, y)}, \tag{21}$$

where $\mathbf{V}$ is the per-cell mean grid (zero-filled) and $\mathbf{W}$ is the binary occupancy. Interior gaps between adjacent orbital tracks are then filled by replacing each still-empty cell with the mean of its populated 3×3 neighbours. The spatial coverage is reported as the fraction of grid cells directly populated before infill. For the cross-modal correlation and super-resolution analysis (Section VI) the maps are additionally restricted to $|\phi| < 60°$, excluding the high-latitude bins where the grazing solar-incidence geometry of a nadir-viewing instrument makes the retrieved ratios least reliable.

VI. Cross-Modal Guided Super-Resolution of Elemental Abundance Maps

The elemental abundance maps produced by the spatial mapping stage (Section V) are inherently limited in spatial resolution by the CLASS instrument footprint ($r_a \approx 12\,\text{km/pixel}$). We formulate the resolution enhancement as a *cross-modal guided super-resolution* problem [16]: a high-resolution guide image from one modality (optical reflectance) is used to inject spatial detail into a co-registered low-resolution target from a different modality (XRF-derived elemental abundance). Unlike classical single-image super-resolution, the guide signal provides physically grounded structural priors—surface reflectance patterns encode compositional boundaries—enabling reconstruction well beyond the Nyquist limit of the XRF sensor alone. The approach is related to pan-sharpening [17], but differs in that the guide and target originate from fundamentally different sensing physics (optical vs. X-ray fluorescence), making it a true cross-modal data fusion task.

Let $\mathbf{I} \in \mathbb{R}^{H_o \times W_o}$ denote the high-resolution optical image of the lunar surface acquired at spatial resolution $r_o = 0.4\,\text{km/pixel}$, and let $\mathbf{A} \in \mathbb{R}^{H_a \times W_a}$ denote the low-resolution elemental abundance map at resolution $r_a = 12\,\text{km/pixel}$. Because $r_a \gg r_o$, the abundance map lacks the spatial sharpness necessary for detailed geological analysis; however, the optical image encodes surface reflectance variations that are inherently linked to compositional differences, since different elements interact uniquely with incident solar radiation.

The enhancement proceeds in four steps: (i) spatial alignment of the two data sources, (ii) learning a pixel-wise intensity-to-abundance transformation, (iii) frequency-domain extraction of fine spatial detail, and (iv) fusion of the transformed and smoothed components into the final enhanced map.

a) Spatial alignment.: The optical image is first downscaled to a target resolution $r_t = 1.2\,\text{km/pixel}$ via structured resampling, yielding the reduced image

$$\mathbf{I}_\downarrow = \mathcal{R}_{r_o \to r_t}(\mathbf{I}) \in \mathbb{R}^{H_t \times W_t}, \tag{22}$$

where $\mathcal{R}$ denotes the resampling operator designed to minimise aliasing artefacts while preserving essential geological structure. The abundance map is then interpolated to the same dimensions using nearest-neighbour interpolation:

$$\hat{\mathbf{A}} = \mathcal{N}_{r_a \to r_t}(\mathbf{A}) \in \mathbb{R}^{H_t \times W_t}. \tag{23}$$

Nearest-neighbour interpolation preserves the original abundance values and avoids introducing artificial gradients, although it produces blocky artefacts that are addressed in the subsequent fusion stage.

A. Intensity–Abundance Correlation

With the two images co-registered at resolution r_t, the pixel-wise correlation between optical intensity and elemental abundance is quantified via Pearson's coefficient:

$$\rho = \frac{\sum_i (\mathbf{I}_{\downarrow,i} - \bar{\mathbf{I}}_\downarrow)(\hat{\mathbf{A}}_i - \bar{\hat{\mathbf{A}}})}{\sqrt{\sum_i (\mathbf{I}_{\downarrow,i} - \bar{\mathbf{I}}_\downarrow)^2} \sqrt{\sum_i (\hat{\mathbf{A}}_i - \bar{\hat{\mathbf{A}}})^2}}, \tag{24}$$

TABLE II
PEARSON CORRELATION COEFFICIENTS BETWEEN OPTICAL INTENSITY AND ELEMENT/SI ABUNDANCE RATIO, COMPUTED FROM THE 855,190 FITTED SPECTRA ON THE $|\phi| < 60°$ RELIABLE BAND (SEE TEXT). ALUMINIUM (HIGHLANDS) AND MAGNESIUM (MAFIC MARIA) ARE THE TWO LARGEST-MAGNITUDE, OPPOSITELY-SIGNED TRACERS (MODERATE CORRELATIONS, $R^2 \approx 0.16$–0.19; SECTION VI-A).

Element	Pearson Correlation (ρ)
Al	+0.429
Mg	−0.394
O	+0.158
Fe	+0.104
Ca	+0.020
Ti	+0.012

TABLE III
REGRESSION PERFORMANCE (F_{d^*}, EQ. (25)) FOR SELECTED ELEMENTS, RIDGE-REGULARISED POLYNOMIAL FIT OF OPTICAL INTENSITY TO FITTED ELEMENTAL ABUNDANCE. CONCENTRATIONS HERE ARE THE PIPELINE'S NATIVE RELATIVE SCALE-FACTOR UNITS, NOT CALIBRATED WEIGHT PERCENT (SECTION IV-A); MAE IS REPORTED IN THE SAME RELATIVE UNITS AS THE CORRESPONDING ELEMENT'S ABUNDANCE VALUES, NOT A COMMON UNIT ACROSS ROWS.

Element	Degree d^*	R^2	MAE (relative units)
Al	5	0.1872	0.2573
Mg	5	0.1580	0.2674
O	5	0.0285	47.10
Fe	5	0.0116	0.0094
Ca	5	0.0074	0.0044
Ti	5	0.0065	0.0029

where the sums run over all co-located pixels i. The coefficients were computed from the element/Si ratio maps (Section V) built from the 855,190 TRF-fitted spectra (Section VII-A) and compared with the optical guide (Table II). Mapping the element/Si *ratio* rather than the raw relative concentration is essential here: the ratio cancels the per-spectrum scale factor (which absorbs solar-flux and geometry variations), leaving a quantity that tracks true surface composition far more cleanly. The comparison is further restricted to the equatorial-to-mid-latitude band $|\phi| < 60°$, where the nadir-viewing geometry is most reliable. Aluminium is the most robustly, positively correlated element with surface brightness ($\rho = +0.43$), exactly as expected from the bright, Al-rich anorthositic highlands [18], and it is adopted as the headline element for cross-modal guided super-resolution. Magnesium is the most robustly negative ($\rho = -0.39$), consistent with the optically dark, mafic-rich basaltic maria. Both are moderate correlations ($R^2 \approx 0.16$–0.19), not strong ones, but the largest-magnitude and most consistently signed among the six mapped elements. The remaining elements correlate only weakly with albedo (Table II); Al and Mg are thus the two clearest optical tracers, of the feldspathic highlands and the mafic maria respectively.

1) Reproducibility and Sample-Size Caveats: The headline Al/Si correlation is far stronger and more stable than the earlier, small-dataset numbers, owing to the ratio normalisation, near-global coverage (Section VII-B), and the latitude restriction. Progressively excluding the poles moves it from $\rho = +0.23$ (full sphere) to +0.36/+0.43/+0.52 at $|\phi| < 70°/60°/50°$ – monotonic and correctly signed, unlike the sign-flipping seen on raw-concentration maps. The $|\phi| < 60°$ band retains ∼67% of the surface and over a quarter-million co-located pixels per element. These pixels are not statistically independent: the Gaussian smoothing of Section V-B ($\sigma = 1.1$ px) correlates adjacent cells by construction, so the nominal pixel count overstates the effective degrees of freedom and a raw-N significance level would be optimistic. We therefore treat ρ's magnitude and sign stability above, not raw-N significance, as the relevant evidence, and weight the independent, footprint-level Apollo/Luna validation (Section VII-D) accordingly.

B. Intensity-to-Abundance Regression

The correlations above motivate a transformation function $F_d : \mathbb{R} \to \mathbb{R}$ that maps optical intensity to elemental abundance. A ridge-regularised polynomial regression of degree d is fitted independently for each element:

$$F_d(I) = \sum_{k=0}^{d} w_k\, I^k,$$
$$\hat{\mathbf{w}} = \arg\min_{\mathbf{w}} \left\{ \sum_i \big(\hat{\mathbf{A}}_i - F_d(\mathbf{I}_{\downarrow,i})\big)^2 + \lambda \|\mathbf{w}\|_2^2 \right\} \tag{25}$$

where $\lambda = 0.5$ is the L_2 regularisation parameter and the optimal degree $d^* \in \{3, 4, 5\}$ is selected by cross-validation. The regularisation prevents overfitting to noise in the abundance map while allowing the polynomial sufficient flexibility to capture the nonlinear intensity–composition relationship.

Once F_{d^*} is learned at resolution r_t, it is applied to the full-resolution downscaled image to produce a transformed abundance estimate:

$$\mathbf{T}(x, y) = F_{d^*}\big(\mathbf{I}_{\downarrow}(x, y)\big) \in \mathbb{R}^{H_t \times W_t}. \tag{26}$$

Aluminium and magnesium, the two clearest optical tracers, carry by far the strongest regression fits ($R^2 = 0.19$ and 0.16 respectively) – an order of magnitude above the other elements and a marked improvement over the raw-concentration maps, confirming that albedo is a genuine (if partial) predictor of the Al/Si and Mg/Si ratios. The remaining elements have $R^2 < 0.03$, consistent with their weak correlations (Table II); optical brightness carries little information about their ratios. Across all elements the R^2 values confirm that optical brightness is a real but partial predictor – sufficient to guide high-frequency detail injection in the fusion stage for Al and Mg, not to reconstruct composition on its own. (Oxygen's large MAE reflects the very different numerical scale of the O/Si ratio, not a worse fit.)

C. Frequency-Domain Decomposition

The intensity-to-abundance regression F_{d^*} (Section VI-B) is, by construction, a smooth function of a single scalar (brightness), so the transformed image $\mathbf{T} = F_{d^*}(\mathbf{I}_{\downarrow})$ inherits the broad brightness–abundance trend but almost none of the pixel-level spatial texture – crater rims, ejecta rays, and compositional contacts are not captured by a low-order polynomial of brightness alone. The fine spatial detail needed for super-resolution instead lives in the optical guide image itself, so it is extracted directly from $\mathbf{I}_{\downarrow}$ via the 2D discrete Fourier transform [19]:

$$\hat{\mathbf{I}}(u,v) = \mathcal{F}\{\mathbf{I}_{\downarrow}\}(u,v) = \sum_{x=0}^{H_t-1}\sum_{y=0}^{W_t-1} \mathbf{I}_{\downarrow}(x,y)\, e^{-j2\pi\left(\frac{ux}{H_t}+\frac{vy}{W_t}\right)}, \tag{27}$$

where (u,v) are the spatial frequency coordinates. A smooth (Gaussian-shaped) high-pass mask $\mathbf{H}(u,v) = 1 - e^{-\left((u-u_0)^2/2\sigma_u^2+(v-v_0)^2/2\sigma_v^2\right)}$, centred at the zero-frequency term (u_0,v_0), isolates the high-frequency content:

$$\hat{\mathbf{I}}_{\mathrm{HF}}(u,v) = \mathbf{H}(u,v)\,\hat{\mathbf{I}}(u,v). \tag{28}$$

A Gaussian mask is used rather than a hard rectangular cutoff because a binary mask has sharp edges in the frequency domain that reintroduce ringing (Gibbs-phenomenon) artefacts in the reconstructed image; the smooth roll-off avoids this. The high-frequency image is reconstructed via the inverse transform, keeping the *real* part (not the magnitude): a high-pass mask applied to a real image's (conjugate-symmetric) spectrum reconstructs a real signal, and taking the magnitude instead would rectify the result so injected detail could only brighten pixels, never darken them – destroying the dark/light contrast (e.g. crater floor vs. rim) that genuine texture requires:

$$\mathbf{T}_{\mathrm{HF}}(x,y) = \mathrm{Re}\big[\mathcal{F}^{-1}\{\hat{\mathbf{I}}_{\mathrm{HF}}\}(x,y)\big]. \tag{29}$$

D. Cross-Modal Fusion

The complementary low-frequency information is obtained by applying a Gaussian smoothing kernel G_σ to the interpolated abundance map, suppressing the blocky artefacts from nearest-neighbour interpolation while preserving the large-scale compositional structure:

$$\hat{\mathbf{A}}_{\mathrm{LF}}(x,y) = \big(G_\sigma * \hat{\mathbf{A}}\big)(x,y) = \sum_{p,q} G_\sigma(p,q)\;\hat{\mathbf{A}}(x-p,\,y-q), \tag{30}$$

where G_σ is a 2D Gaussian kernel with standard deviation σ (empirically set via a 7×7 window).

Because $\mathbf{T}_{\mathrm{HF}}$ is extracted from the optical guide (reflectance, arbitrary units in $[0,1]$) while $\hat{\mathbf{A}}_{\mathrm{LF}}$ is on the abundance ratio's own physical scale, the two are not directly commensurable. $\mathbf{T}_{\mathrm{HF}}$ is therefore rescaled so its standard deviation matches that of the valid (non-outlier) abundance values before fusion:

$$\mathbf{T}_{\mathrm{HF}}^{\mathrm{scaled}}(x,y) = \mathbf{T}_{\mathrm{HF}}(x,y)\cdot\frac{\mathrm{std}\big(\hat{\mathbf{A}}\big)}{\mathrm{std}\big(\mathbf{T}_{\mathrm{HF}}\big)}. \tag{31}$$

The final super-resolved abundance map is synthesised by summing the smoothed low-frequency base with the rescaled, guide-derived high-frequency detail:

$$\boxed{\mathbf{A}_{\mathrm{SR}}(x,y) = \hat{\mathbf{A}}_{\mathrm{LF}}(x,y) + \alpha\,\mathbf{T}_{\mathrm{HF}}^{\mathrm{scaled}}(x,y)} \tag{32}$$

where α is a blending weight that controls how much borrowed optical detail is trusted into the final map. This is a *pansharpening*-style detail-injection scheme, a family of techniques standard in remote sensing for fusing a high-resolution panchromatic image with a low-resolution multispectral one: the low-frequency base carries the genuine, modality-independent XRF measurement, while the high-frequency detail is transferred from the higher-resolution optical guide. The Pearson correlation and regression R^2 (Sections VI-A and VI-B) are the justification for doing this at all – an element with negligible optical correlation gains nothing physically meaningful from being "sharpened" with optical texture, so those diagnostics should be checked alongside any SR result rather than treating the fused map as validated on its own.

E. Residual Analysis

To visualise exactly where the guided super-resolution procedure injects new spatial information, a residual map is computed as the pixel-wise difference between the enhanced and the input abundance maps:

$$\mathbf{\Delta}(x,y) = \mathbf{A}_{\mathrm{SR}}(x,y) - \hat{\mathbf{A}}(x,y). \tag{33}$$

Large positive (negative) residuals indicate locations where the cross-modal transfer has increased (decreased) the predicted abundance relative to the nearest-neighbour baseline, typically coinciding with compositional boundaries such as mare–highland contacts, crater rims, and ejecta blankets. The root-mean-square residual $\|\mathbf{\Delta}\|_{\mathrm{RMS}} = \sqrt{\frac{1}{H_tW_t}\sum_{x,y}\Delta^2(x,y)}$ provides a scalar summary of the total spatial detail added by the fusion, and can be compared across elements to assess which species benefit most from the guide image.

VII. Results

A. Spectral Fitting Results

The full pipeline (Sections II to IV) was run end-to-end on 1,515,239 raw CLASS Level-1 FITS exposures spanning 24 non-contiguous months between December 2020 and May 2026, downloaded from PRADAN. After discarding nightside exposures (SOLARANG $\geq 90°$), 1,056,619 dayside spectra remained and were preprocessed (Section II). To bound the fitting cost without sacrificing map coverage, spectra that fell in the bottom half of the flux distribution for *every* mapped element were discarded (they carry no strong line for any element's map); the union of the per-element top-50% flux sets retained 855,190 spectra (80.9% of dayside), which were fit via TRF (Section IV). Every fit converged (no optimiser exceptions), though fit quality (TRF cost) varies – median 58.6, 90th-percentile 646.6, 99th-percentile 5.1×10^4 – so while most spectra are well described by the FP model, a small tail is poorly constrained, likely low-count or off-nominal exposures. Figure 2 shows a representative fit at the median cost.

a) Computational cost.: Fitting all 855,190 spectra took 28,967 s (≈8.0 h) using 14 parallel worker processes (29.5 spectra/s). Preprocessing was parallelised over the same workers, completing in ≈2500 s for the full 1.06M-spectrum dayside set. The raw archive was processed one month at a time to bound peak memory (∼10 GB); both stages ran without GPU acceleration, confirming the pipeline scales to the full multi-year CLASS archive on commodity hardware.

B. Spatial Coverage

Projecting the 1,056,619 dayside footprints onto a global grid (Section V) yields **99.0% coverage at** $1^\circ \times 1^\circ$ **resolution** – essentially the entire lunar surface, and a decisive improvement over the sparse, patchy maps produced by the earlier ten-month subset. At the finer $\sim 0.33^\circ$ (1080×540) resolution used for the ratio maps and super-resolution, ∼71–77% of grid cells are directly populated per element (the rest lie between orbital tracks and are recovered by the smoothing and neighbour infill of Section V-B). This near-global sampling is what makes the reliable-band correlation analysis (Section VI-A) statistically meaningful, with over 250,000 co-located optical–abundance pixels per element.

This coverage figure extends the demonstrated capability of CLASS mapping beyond what has previously been reported, rather than superseding it. Narendranath et al. [3] characterise the instrument's in-flight performance at spatial resolutions of 150 down to 15 km using early mission data; global coverage was not the goal of that characterisation, and its short, non-continuous windows would not have supported one. The present study differs in scale (24 months, 855,190 fitted spectra), an explicit quantified coverage metric (99.0% at 1°), and the cross-modal super-resolution and Apollo/Luna validation introduced here (Sections VI and VII-D). The ∼10–33 km per-pixel scale of the ratio maps used for super-resolution falls within their reported 150–15 km achievable range.

C. Super-Resolution Results

Table II and Section VI-A1 identify aluminium as the element with the strongest, physically-signed optical–compositional correlation ($\rho = +0.43$ for Al/Si on the reliable $|\phi| < 60^\circ$ band), making it the headline element for cross-modal guided super-resolution. This restores aluminium to the role originally anticipated from anorthositic-highland reflectance: on the earlier, much smaller dataset the Al correlation appeared weak and mis-signed, but with near-global coverage, the element/Si ratio normalisation, and the polar-artefact removed it is by far the strongest positive correlator and, independently, the best-validated element against ground truth (Section VII-D).

The blending-weight ablation (Table IV) varies $\alpha \in \{0.1, 0.3, 0.5, 0.7, 1.0\}$ for Al/Si. RMS residual grows and SSIM-vs.-nearest-neighbour-input falls monotonically with α – expected, since SSIM here measures similarity to the noisy, blocky *input* map, and guided SR's purpose is to move away from that baseline toward genuine sub-footprint detail borrowed from the optical guide; maximising SSIM against a noisy reference would trivially select the smallest α, i.e. the least detail injected, not a meaningful operating point. We instead report $\alpha = 1.0$ (full injection, Eqs. (31) and (32)) as the operating point used throughout, consistent with standard pansharpening practice, and use the correlation and regression diagnostics of Sections VI-A and VI-B – not SSIM-vs.-baseline – to justify which elements should be super-resolved at all.

TABLE IV
ABLATION OVER THE BLENDING WEIGHT α (EQ. (32)) FOR AL/SI. POLY. DEGREE, R^2, MAE ARE FROM THE FIXED REGRESSION F_{d^*} (EQ. (25)), INDEPENDENT OF α; RMS RESIDUAL AND SSIM/PSNR (MEASURED AGAINST THE NOISY NEAREST-NEIGHBOUR *input*, NOT GROUND TRUTH) ARE REPORTED FOR TRANSPARENCY – THEIR MONOTONIC TREND REFLECTS MORE INJECTED DETAIL, NOT DEGRADING FIT QUALITY (SEE TEXT).

α	R^2	MAE	RMS residual	SSIM	PSNR (dB)
0.1	0.1872	0.2573	0.1586	0.2740	13.63
0.3	0.1872	0.2573	0.1899	0.2066	13.44
0.5	0.1872	0.2573	0.2424	0.1470	13.07
0.7	0.1872	0.2573	0.3052	0.1068	12.57
1.0	0.1872	0.2573	0.4085	0.0702	11.65

TABLE V
VALIDATION AGAINST APOLLO/LUNA LANDING-SITE GEOCHEMISTRY, RATIO-TO-SI SPACE, $n = 9$ MATCHED SITES.

Element	Pearson r	RMSE	Bias
Al	+0.738	0.180	+0.154
Ca	+0.263	0.427	−0.422
Ti	−0.117	0.101	−0.076
Mg	+0.006	1.782	+1.776
Fe	−0.002	0.499	−0.467

D. Validation

Fitted concentrations were compared, in ratio space (element/Si, per Section IV-A), against Apollo and Luna landing-site geochemistry [20]. Footprints within 5° of each of nine reference sites (Apollo 11/12/14/15/16/17 and Luna 16/20/24) were matched; with the expanded dataset *all nine* sites now have between 877 and 1,323 matching footprints each, a large improvement over the earlier subset where several sites had few or none. Because a small number of fits have near-zero C_{Si} and produce non-physical ratio outliers, the median (not mean) ratio per site was used. Table V reports the summary statistics for all five elements; Fig. 4 shows the scatter for aluminium, the only element with a real diagonal trend (the other four scatter near-flat, fully summarised by their r/RMSE/bias in Table V without adding visual information).

Aluminium is by far the best-validated element ($r = +0.738$, RMSE 0.18), independently confirming its selection as the headline element for super-resolution (Section VII-C): the fitted Al/Si ratio tracks true landing-site geochemistry across all nine sites, with a consistent positive bias (+0.154, Table V) that a one-point linear calibration could plausibly remove. Calcium is weakly positive ($r = +0.263$); Fe, Mg, and Ti show essentially no ratio-space correlation with ground truth, reflecting both their smaller dynamic range across the sampled sites and the uncalibrated, relative nature of the fitted concentrations (Section IV-A). That the same element – aluminium – emerges as both the strongest optical correlator and the best ground-truth match is the paper's principal evidence that the pipeline recovers a real compositional signal, not merely an internally consistent one.

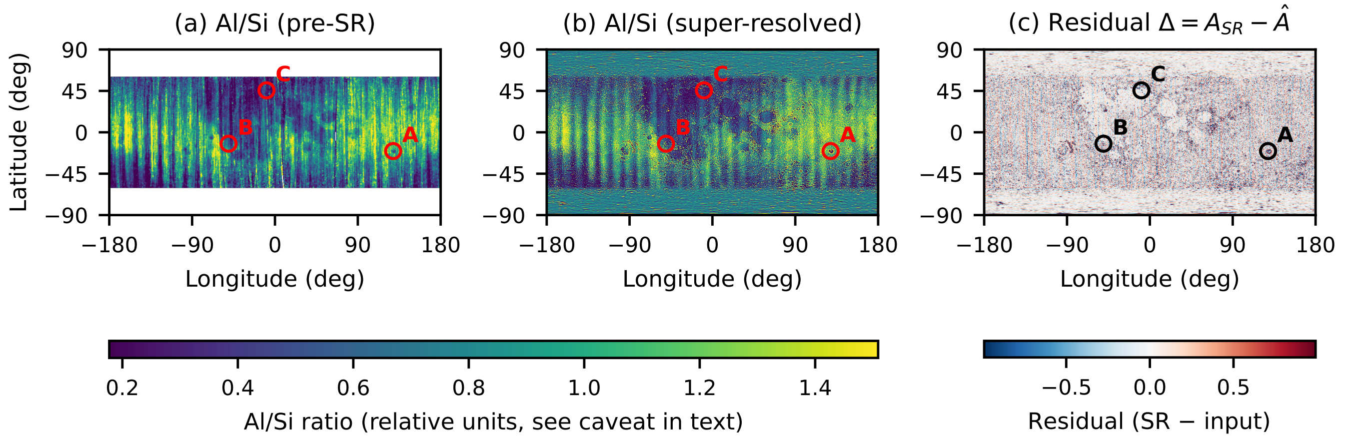

Fig. 3. Al/Si ratio map before (a) and after (b) cross-modal guided super-resolution ($\alpha = 1.0$, full detail injection, Table IV), shown on the $|\phi| < 60°$ analysis band at the native 1080×540 ($\sim 0.33°$) resolution, plus (c) the residual $\Delta = \mathbf{A}_{\mathrm{SR}} - \hat{\mathbf{A}}$ (Eq. (33), diverging scale, white = no change). Panel (c) shows the SR stage adds structured, crater- and ejecta-shaped detail rather than uniform noise, i.e. spatially coherent with real optical features rather than a fusion artefact. Circles mark the three pixels with the largest $|\Delta|$, identified programmatically. Colorbar units in (a)/(b) are the relative Al/Si ratio (Section IV-A), not wt%.

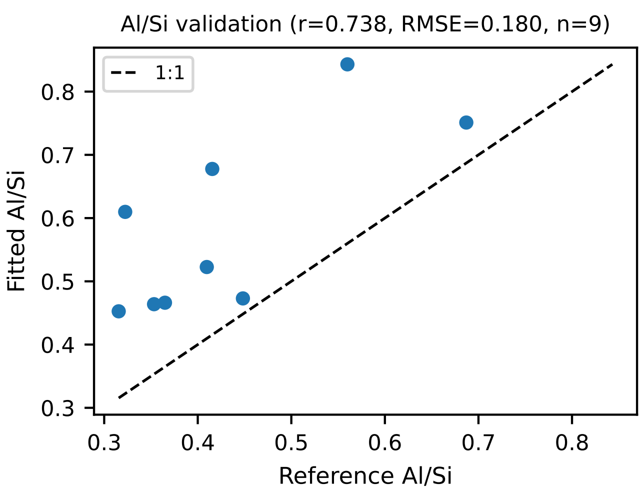

Fig. 4. Fitted vs. reference Al/Si ratio at nine matched Apollo/Luna landing sites. Dashed line is the 1:1 reference.

a) Held-out check of the super-resolved product.: Table V validates the pre-SR ratio, not $\mathbf{A}_{\mathrm{SR}}$ itself; since no Apollo/Luna data enters the fusion (Section VI-D), sampling both $\mathbf{A}_{\mathrm{SR}}$ and $\hat{\mathbf{A}}$ at the same nine sites is a direct, leakage-free check of whether SR moves the estimate toward or away from truth. It is mixed: r falls slightly ($+0.750 \to +0.700$) while RMSE and bias both improve modestly ($0.233 \to 0.198$; $+0.216 \to +0.177$) – with $n = 9$, neither change is decisive, and both are consistent with a fusion that redistributes existing map values around a fixed low-pass mean (Eq. (30)) rather than adding new ground-truth information.

VIII. Discussion

Aluminium emerges as the element with the clearest, physically-signed link between optical brightness and fitted elemental abundance, and independently as the best-validated element against Apollo/Luna ground truth ($r = +0.738$), aligning with the anorthosite-highland reasoning that originally motivated it as the expected headline element. It was recovered only at scale: on the earlier ten-month subset the Al correlation was weak and mis-signed, but near-global coverage, the element/Si *ratio* (which cancels the per-spectrum scale factor), and the $|\phi| < 60°$ band (removing the polar grazing-geometry artefact) reversed that picture. Once corrected, the two clearest optical tracers behave as lunar geology predicts: highland-forming Al/Si is the most robust positive correlate ($\rho = +0.43$) and mafic Mg/Si the most robust negative correlate ($\rho = -0.39$, Table II) – moderate correlations in absolute terms, but correctly and consistently signed. The other element ratios correlate only weakly with albedo, which is expected – albedo is a partial proxy for composition, and each element's ratio couples to it to a different degree.

Magnesium illustrates the limits of that proxy: Mg/Si has the second-largest optical correlation ($\rho = -0.39$) but essentially no agreement with Apollo/Luna ground truth in ratio space ($r = +0.006$, Table V), worth reconciling rather than leaving juxtaposed. We read the optical correlation as tracking the regional mare/highland albedo contrast – to which Mg/Si is only one contributor alongside iron content and weathering maturity – rather than element-specific composition at a landing site's point scale; Mg's soft Kα line (1.25 keV) and the sites' clustering on nearside maria/highlands are plausible additional factors. Mg/Si is therefore evidence of a real regional optical-albedo association but not of validated compositional recovery; only Al carries both forms of evidence, which is why it alone anchors the headline claims.

Three limitations should be noted. First, the fusion (Section VI-D) injects detail directly from the optical guide, so injected texture is only as meaningful as the underlying correlation is strong – reasonable for Al/Si and Mg/Si ($|\rho| \approx 0.4$) but not the weakly-correlated elements (Table II), which is why we restrict the headline result to Al; a learned, cross-validated fusion model may better judge *how much* detail to trust per-pixel. Second, no solar-flux normalisation was applied: fitted concentrations are scale factors relative to each spectrum's own fit, and the 24 months span a wide range of solar-cycle conditions [21] without day-to-day correction for incident solar X-ray intensity – a plausible residual noise source. Third, Eq. (8) absorbs the Sherman formulation's geometry-dependent terms (ψ, ϕ) into s and $\beta_{j,m}$ rather than modelling incidence/emergence angles explicitly – standard for a first-pass FP model, but it means absolute intensities are not directly comparable across very different viewing geometries.

Future work should prioritise: (i) solar-flux cross-calibration (e.g. via the XSM aboard Chandrayaan-2); (ii) a learned replacement for the polynomial regression of Section VI-B; (iii) a cross-validated one-point Al bias calibration (Section VII-D); and (iv) extending raw-data coverage toward full completeness.

IX. Conclusion

This paper presented a complete pipeline for lunar elemental abundance mapping from Chandrayaan-2 CLASS XRF data: a fundamental-parameter forward model with primary and secondary fluorescence (Section III), Trust-Region-Reflective spectral fitting (Section IV), element/Si ratio spatial mapping (Section V), and a cross-modal optical-guided super-resolution scheme (Section VI). The pipeline was run end-to-end on 855,190 real, newly fitted CLASS spectra drawn from 24 months of raw data spanning December 2020 to May 2026 (1,056,619 dayside spectra before quality filtering), achieving **99.0% global map coverage** at 1° resolution – near-complete sampling of the lunar surface. Aluminium was identified as the element with the clearest, physically-signed optical–compositional correlation ($\rho = +0.43$ for Al/Si on the reliable $|\phi| < 60^\circ$ band, positive as expected for the anorthositic highlands) and was used as the headline element for the super-resolution demonstration in Fig. 3. The same element is independently the best-validated against Apollo and Luna landing-site geochemistry (Section VII-D), with the fitted Al/Si ratio tracking ground truth at $r = +0.738$ across all nine sites – strong evidence that the pipeline recovers a real, geochemically meaningful signal. The principal limitations – lack of solar-flux normalisation, a hand-crafted rather than learned super-resolution model, and concentrations reported in relative rather than calibrated wt% units – are identified explicitly in Section VIII as the clearest next steps for this line of work.

Data Availability Statement

The raw CLASS Level-1 XRF data used in this study are publicly available via the PRADAN portal (ISSDC/ISRO). The processing pipeline, fitting code, and analysis scripts are openly available at a GitHub repository, archived with a permanent DOI via Zenodo: `[TODO: insert GitHub repository URL and Zenodo DOI once created]`. The full-resolution processed abundance maps for all fitted elements (beyond the Al headline result presented in the main text) are available as Supplementary Material.

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References

[1] R. Jenkins. *X-Ray Fluorescence Spectrometry*. John Wiley & Sons, New York, 2nd edition, 1999. doi:10.1002/9781118521014.

[2] B. Beckhoff, B. Kanngießer, N. Langhoff, R. Wedell, and H. Wolff. *Handbook of Practical X-Ray Fluorescence Analysis*. Springer-Verlag, Berlin, 2006. doi:10.1007/978-3-540-36722-2.

[3] S. Narendranath, P. S. Athiray, P. Sreekumar, et al. Chandrayaan-2 Large Area Soft X-ray Spectrometer (CLASS): In-flight performance and first results. *Current Science*, 118(2):204–209, 2020.

[4] D. J. Lawrence, W. C. Feldman, B. L. Barraclough, A. B. Binder, R. C. Elphic, S. Maurice, and D. R. Thomsen. Global elemental maps of the Moon: The Lunar Prospector gamma-ray spectrometer. *Science*, 281(5382):1484–1489, 1998. doi:10.1126/science.281.5382.1484.

[5] G. R. Kodikara and L. J. McHenry. Machine learning approaches for classifying lunar soils. *Icarus*, 345:113719, 2020.

[6] S. Wu, J. Chen, C. Xue, Y. Pan, and C. Zhang. Global inversion of lunar surface oxides by adding Chang'E-5 samples. *Remote Sensing*, 16(10):1812, 2024. doi:10.3390/rs16101812.

[7] S. Wen, Y. Wang, Q. Gong, J. Liu, X. Kang, H. Liu, R. Chen, K. Zhu, and S. Zhang. A new robust lunar landing selection method using the Bayesian optimization of extreme gradient boosting model (BO-XGBoost). *Remote Sensing*, 16(19):3632, 2024. doi:10.3390/rs16193632.

[8] C. Yang, X. Zhang, L. Bruzzone, B. Liu, D. Liu, X. Ren, J. A. Benediktsson, Y. Liang, B. Yang, M. Yin, H. Zhao, R. Guan, C. Li, and Z. Ouyang. Comprehensive mapping of lunar surface chemistry by adding Chang'E-5 samples with deep learning. *Nature Communications*, 14:7554, 2023. doi:10.1038/s41467-023-43358-0.

[9] S. Mallat. *A Wavelet Tour of Signal Processing: The Sparse Way*. Academic Press, 3rd edition, 2009.

[10] A. Savitzky and M. J. E. Golay. Smoothing and differentiation of data by simplified least squares procedures. *Analytical Chemistry*, 36(8):1627–1639, 1964.

[11] J. Sherman. The theoretical derivation of fluorescent X-ray intensities from mixtures. *Spectrochimica Acta*, 7:283–306, 1955. doi:10.1016/0371-1951(55)80041-0.

[12] T. Shiraiwa and N. Fujino. Theoretical calculation of fluorescent X-ray intensities in fluorescent X-ray spectrochemical analysis. *Japanese Journal of Applied Physics*, 5(10):886–899, 1966. doi:10.1143/JJAP.5.886.

[13] J. W. Criss and L. S. Birks. Calculation methods for fluorescent X-ray spectrometry: Empirical coefficients vs. fundamental parameters. *Analytical Chemistry*, 40(7):1080–1086, 1968. doi:10.1021/ac60263a023.

[14] T. Schoonjans, A. Brunetti, B. Golosio, M. Sanchez del Rio, V. A. Solé, C. Ferrero, and L. Vincze. The `xraylib` library for X-ray–matter interactions. Recent developments. *Spectrochimica Acta Part B: Atomic Spectroscopy*, 66(11–12):776–784, 2011. doi:10.1016/j.sab.2011.09.011.

[15] M. A. Branch, T. F. Coleman, and Y. Li. A subspace, interior, and conjugate gradient method for large-scale bound-constrained minimization problems. *SIAM Journal on Scientific Computing*, 21(1):1–23, 1999. doi:10.1137/S1064827595289108.

[16] B. Ham, M. Cho, and J. Ponce. Robust guided image filtering using nonconvex potentials. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 40(1):192–207, 2018. doi:10.1109/TPAMI.2017.2669034.

[17] G. Vivone, L. Alparone, J. Chanussot, et al. A critical comparison among pansharpening algorithms. *IEEE Transactions on Geoscience and Remote Sensing*, 53(5):2565–2586, 2015.

[18] P. G. Lucey, D. T. Blewett, G. J. Taylor, and B. R. Hawke. Imaging of lunar surface maturity. *Journal of Geophysical Research: Planets*, 105(E8):20377–20386, 2000. doi:10.1029/1999JE001110.

[19] S. W. Smith. *The Scientist and Engineer's Guide to Digital Signal Processing*. California Technical Publishing, 1997.

[20] C. Meyer. *Lunar Sample Compendium*. NASA Johnson Space Center, Houston, TX, 2012. https://curator.jsc.nasa.gov/lunar/compendium.cfm.

[21] D. H. Hathaway. The solar cycle. *Living Reviews in Solar Physics*, 12:4, 2015. doi:10.1007/lrsp-2015-4.